

Problem

Given that $v_c(0) = 2 \cos(155^\circ)$ V, what is the instantaneous voltage across a $2\text{-}\mu\text{F}$ capacitor when the current through it is $i = 4 \sin(10^6 t + 25^\circ)$ A?

Step-by-step solution

Step 1 of 3

Refer to section 9.4 in the textbook.

Write the expression for the instantaneous voltage across capacitor.

$$v_c(t) = \frac{1}{C} \int i dt$$

Substitute $4 \sin(10^6 t + 25^\circ)$ A for i , and $2\text{ }\mu\text{F}$ for C .

$$\begin{aligned} v_c(t) &= \frac{1}{(2 \times 10^{-6})} \int (4 \sin(10^6 t + 25^\circ)) dt \\ &= \frac{4}{(2 \times 10^{-6})} \left[\int \sin(10^6 t + 25^\circ) dt \right] \\ &= \frac{4}{(2 \times 10^{-6})} \left[\frac{-\cos(10^6 t + 25^\circ)}{10^6} \right] + c \end{aligned}$$

$$v_c(t) = -2 \cos(10^6 t + 25^\circ) + c \dots\dots (1)$$

Here,

c is a constant of integration, and its value are determined using initial conditions.

Step 2 of 3

Determine the value of c using initial conditions at $t = 0$.

Substitute 0 for t , and $v_c(0) = 2 \cos(155^\circ)$ V in equation (1).

$$2 \cos(155^\circ) = -2 \cos(10^6(0) + 25^\circ) + c$$

$$c = 2 \cos(155^\circ) + 2 \cos(25^\circ)$$

$$c = 2((-0.906) + (0.906))$$

$$c = 0$$

Step 3 of 3

Substitute 0 for ϵ in equation (1).

$$\begin{aligned}v_C(t) &= -2 \cos(10^6 t + 25^\circ) + 0 \\&= -2 \cos(10^6 t + 25^\circ)\end{aligned}$$

Consider $\cos \theta = \sin(90^\circ - \theta)$.

$$\begin{aligned}v_C(t) &= -2 \sin(90^\circ - (10^6 t + 25^\circ)) \\&= -2 \sin(-10^6 t + 65^\circ) \\&= -2 \sin(-(10^6 t - 65^\circ))\end{aligned}$$

Consider $\sin(-\theta) = -\sin(\theta)$.

$$v_C(t) = 2 \sin(10^6 t - 65^\circ) \text{ V}$$

Therefore, the instantaneous voltage across capacitor $v_C(t)$ is $\boxed{2 \sin(10^6 t - 65^\circ) \text{ V}}$.